

Conjugate of a Complex Number

Rigorous Algebraic Foundations, Geometric Interpretations, and
Comprehensive Explanations of Core Properties

2.4 Formal Definition

Introducing the Conjugate Operator

In mathematics, every complex number has a unique symmetric counterpart called its **conjugate**. Taking the conjugate of a complex number algebraically mirrors it across the real coordinate plane.

- ☞ Constructed by reversing the sign of the imaginary coefficient.
- ☞ Denoted explicitly by placing a horizontal bar (vinculum) over the variable as $\bar{z}$.
- ☞ Crucial in simplifying ratios of complex values.

Definition 2.3

"The conjugate of the complex number $x + iy$ is defined as the complex number $x - iy$."

$$\text{If } z = x + iy, \text{ then } \bar{z} = x - iy$$

Algebraic Operations & Real Products

When any complex value z is multiplied by its own conjugate $\bar{z}$, the imaginary components mathematically cancel out. This produces a purely real number equal to the sum of the squares of its coefficients.

The Algebraic Transition

$$(x + iy)(x - iy) = x^2 - (iy)^2$$

Since $i^2 = -1$, the subtraction becomes addition:

$$= x^2 + y^2$$

Example (i)

$$\text{Let } z = 2 - 5i$$

To obtain the conjugate, flip the coefficient of i :

$$\bar{z} = 2 + 5i$$

Example (ii)

Compute the product of $1 + 3i$ and its conjugate:

$$(1 + 3i)(1 - 3i) = (1)^2 - (3i)^2$$

$$= 1 - (-9) = 10$$

2.4.1 Geometrical Representation

Symmetry in the Argand Plane

Geometrically, the complex conjugate of z is represented as the **reflection of z across the real axis (horizontal axis)** in the Argand plane.

- ☞ The real coordinate x remains completely unchanged.
- ☞ The imaginary coordinate y flips across the horizontal line.
- ☞ This maps the coordinate point (x, y) directly onto $(x, -y)$.

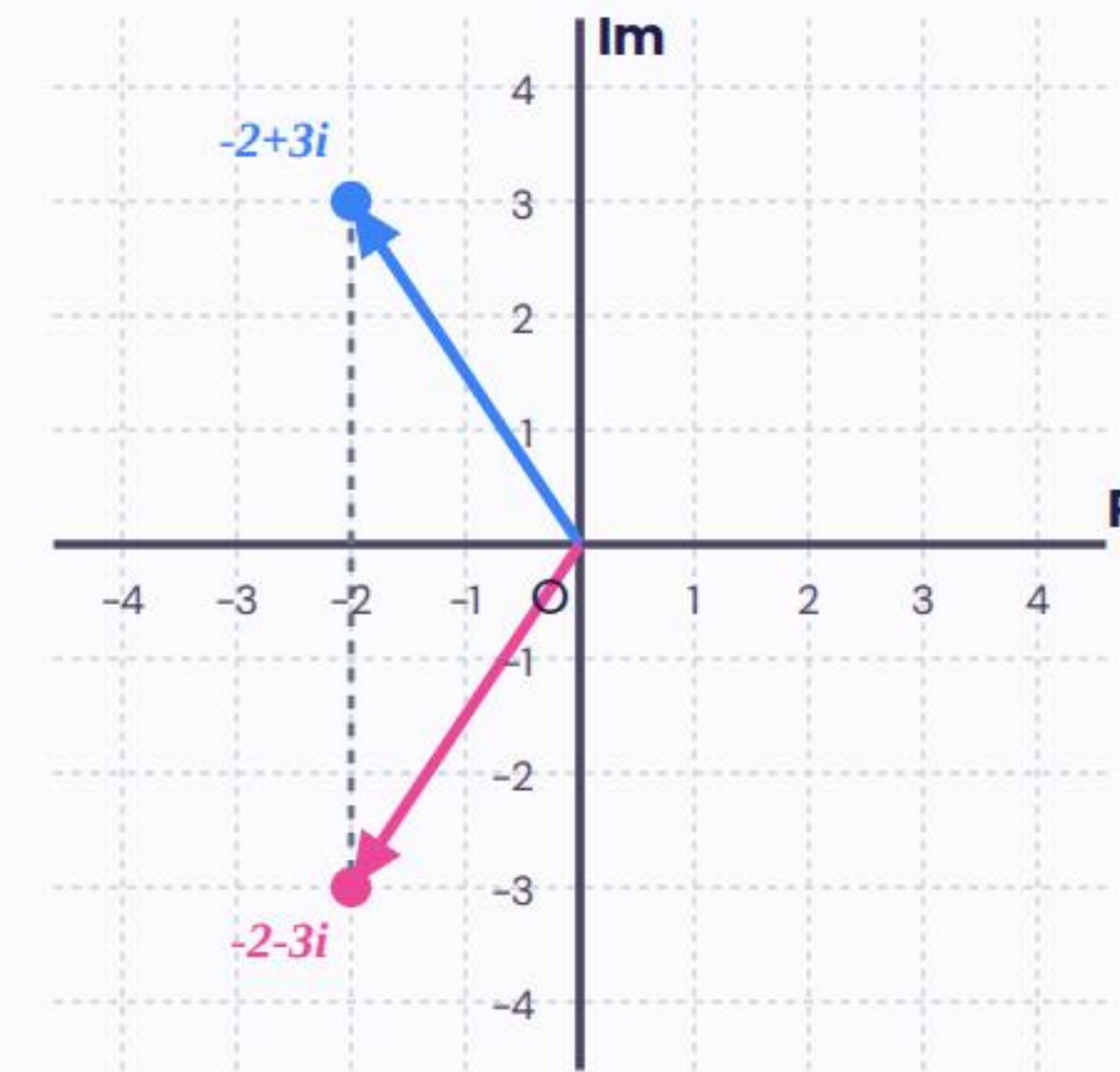


Fig 2.14: Conjugation of $-2 + 3i$

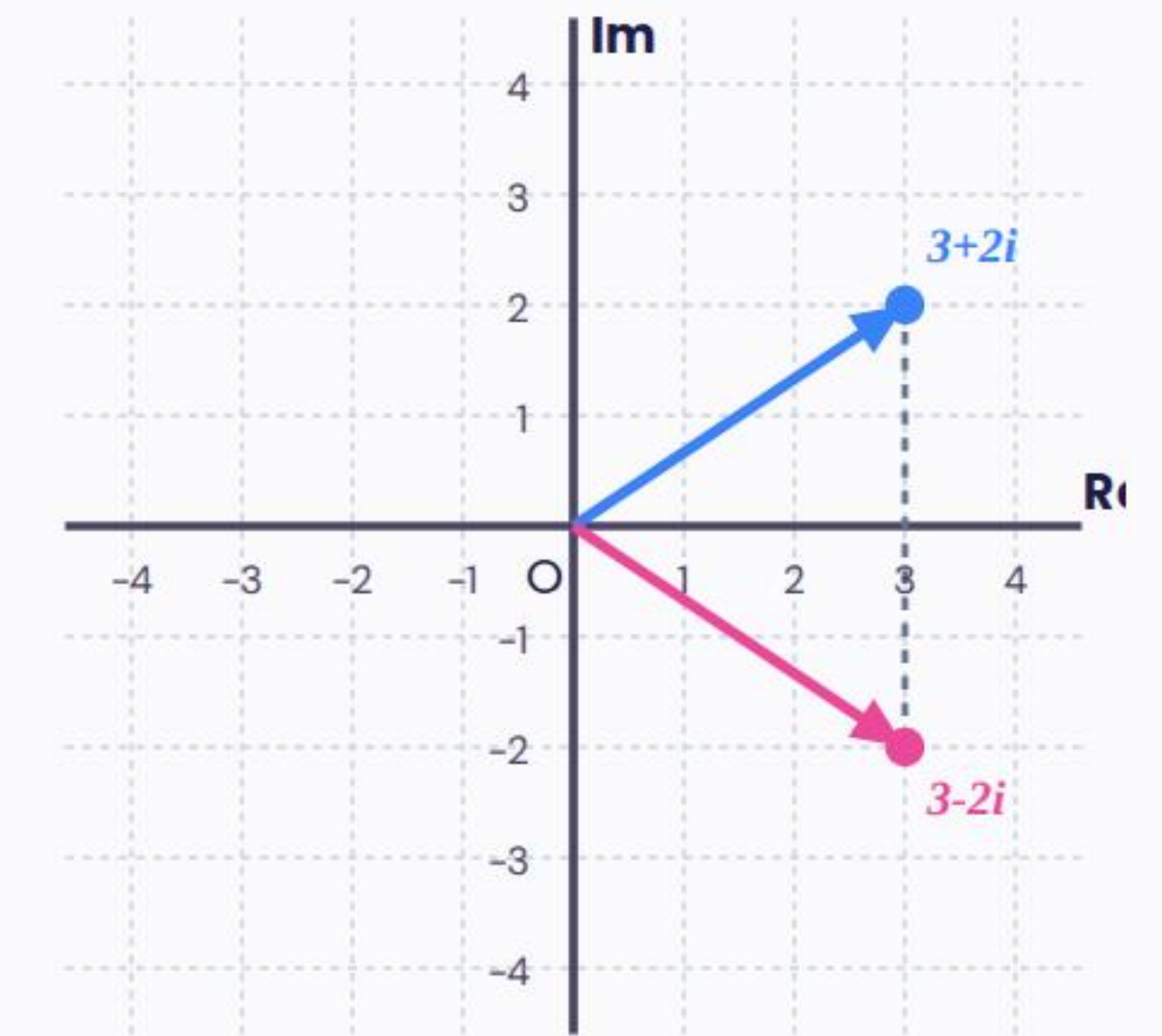


Fig 2.15: Conjugation of $3 + 2i$

Rationalization & Complex Division

Why is the Conjugate Useful?

We cannot directly divide complex numbers when an imaginary component exists in the denominator. The conjugate acts as a "realizing factor" to eliminate imaginary terms.

Rationalization Rule

To divide two complex numbers, replace the complex number in the denominator with a real number by multiplying both numerator and denominator by the conjugate of the denominator.

The General Process

Let the quotient be defined as:

$$z_1 / z_2$$

Multiply with conjugate of denominator:

$$(z_1 / z_2) \cdot (\overline{z_2} / \overline{z_2}) = (z_1 \overline{z_2}) / |z_2|^2$$

2.4.2 Properties of Complex Conjugates

Conjugates behave predictably under linear and arithmetic transformations. Below are the first 5 fundamental algebraic properties:

1

Conjugate of a Sum

$$\overline{z_1 + z_2} = \overline{z_1} + \overline{z_2}$$

2

Conjugate of a Difference

$$\overline{z_1 - z_2} = \overline{z_1} - \overline{z_2}$$

3

Conjugate of a Product

$$\overline{z_1 z_2} = \overline{z_1} \overline{z_2}$$

4

Conjugate of a Quotient

$$\overline{(z_1/z_2)} = \overline{z_1}/\overline{z_2}$$

5

Real Part Formula

$$\operatorname{Re}(z) = (z + \overline{z}) / 2$$

Properties of Complex Conjugates (6–10)

The remaining properties define the relationships between purely real, purely imaginary, and exponents of conjugates:

6

Imaginary Part Formula

$$\text{Im}(z) = (z - \bar{z}) / 2i$$

7

Integer Powers

$$\overline{z^n} = (\bar{z})^n$$

8

Real Number Condition

$$z = \bar{z} \iff z \text{ is Real}$$

9

Imaginary Condition

$$z = -\bar{z} \iff z \text{ is Purely Im}$$

10

Double Conjugation Involution

$$\overline{\bar{z}} = z$$

Proof Spotlight: Property 1

Conjugate of a Sum is the Sum of Conjugates

We establish algebraic equivalence by representing the values in their standard binomial real-imaginary components:

"The addition of two values followed by conjugation yields the same result as taking individual conjugates and then summing them."

Generalization: By mathematical induction, this property scales perfectly to any finite sum of terms:

$$\overline{z_1 + z_2 + \dots + z_n} = \overline{z_1} + \overline{z_2} + \dots + \overline{z_n}$$

Rigorous Algebraic Proof

Let $z_1 = x_1 + iy_1$ and $z_2 = x_2 + iy_2$.

Step 1: Compute Left Hand Side (LHS)

$$\begin{aligned} z_1 + z_2 &= (x_1 + x_2) + i(y_1 + y_2) \\ \overline{z_1 + z_2} &= (x_1 + x_2) - i(y_1 + y_2) \end{aligned}$$

Step 2: Compute Right Hand Side (RHS)

$$\begin{aligned} \overline{z_1} + \overline{z_2} &= (x_1 - iy_1) + (x_2 - iy_2) \\ &= (x_1 + x_2) - i(y_1 + y_2) \end{aligned}$$

✓ LHS = RHS. Q.E.D.

Proof Spotlight: Property 3

Conjugate of a Product is Product of Conjugates

We establish that multiplication and conjugation operations commute. Let's expand both sides systematically using standard binomial expansions.

"Taking the complex product first, then computing the conjugate, gives an equivalent result to applying the conjugation first."

Rigorous Proof

Let $z^1 = x^1 + iy^1$ and $z^2 = x^2 + iy^2$.

LHS Calculation:

$$\begin{aligned} z^1 z^2 &= (x^1 x^2 - y^1 y^2) + i(x^1 y^2 + x^2 y^1) \\ \overline{z^1 z^2} &= (x^1 x^2 - y^1 y^2) - i(x^1 y^2 + x^2 y^1) \end{aligned}$$

RHS Calculation:

$$\begin{aligned} \overline{z^1} \overline{z^2} &= (x^1 - iy^1)(x^2 - iy^2) \\ &= x^1 x^2 + i^2 y^1 y^2 - i(x^1 y^2 + x^2 y^1) \\ &= (x^1 x^2 - y^1 y^2) - i(x^1 y^2 + x^2 y^1) \end{aligned}$$

✓ LHS = RHS. Q.E.D.

Proof Spotlight: Property 9

Condition for Purely Imaginary Numbers

We mathematically verify the property that a complex number is purely imaginary if and only if it is equal to the negative of its conjugate.

"An identity where the real part is zero, leaving only the imaginary coordinate on the vertical axis."

Biconditional Verification

Let $Z = x + iy$. Its conjugate is $\bar{Z} = x - iy$.

Assume that:

$$Z = -\bar{Z}$$

Substitute algebraic form:

$$\begin{aligned} x + iy &= -(x - iy) \\ x + iy &= -x + iy \end{aligned}$$

Simplify by subtracting iy from both sides:

$$2x = 0 \Rightarrow x = 0$$

✔ Since $x = 0$, z is purely imaginary. Proved.

Solved Lecture Examples

EXAMPLE 1

Find conjugate of $z = 2/(1 - i)$

First, convert to standard form by multiplying with the denominator's conjugate:

$$z = (2/(1 - i)) \cdot ((1 + i)/(1 + i))$$

$$z = 2(1 + i)/2 = 1 + i$$

Apply conjugate rule:

$$\bar{z} = 1 - i$$

EXAMPLE 2

Find conjugate of $z = (3 + 4i)(1 - 2i)$

Using Property 3, distribute the conjugation first, avoiding long multiplication:

$$\bar{z} = \overline{(3 + 4i)} \cdot \overline{(1 - 2i)}$$

$$\bar{z} = (3 - 4i)(1 + 2i)$$

Now, expand the simplified conjugates:

$$= 3 + 6i - 4i - 8i^2$$

$$= 11 + 2i$$



Thank You

Are there any questions on the properties, mathematical induction, or geometric representations of complex conjugates?

XII - MATHEMATICS